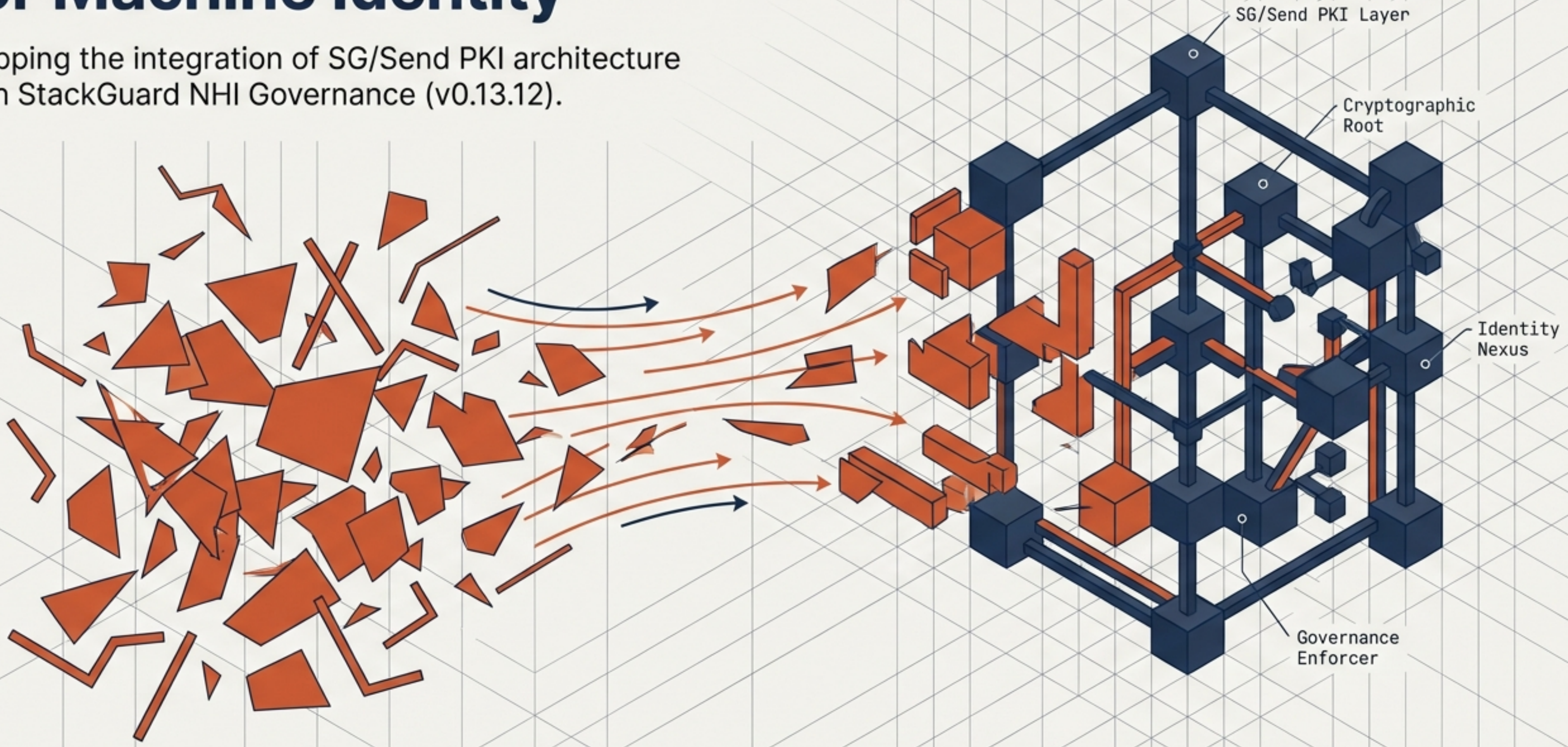
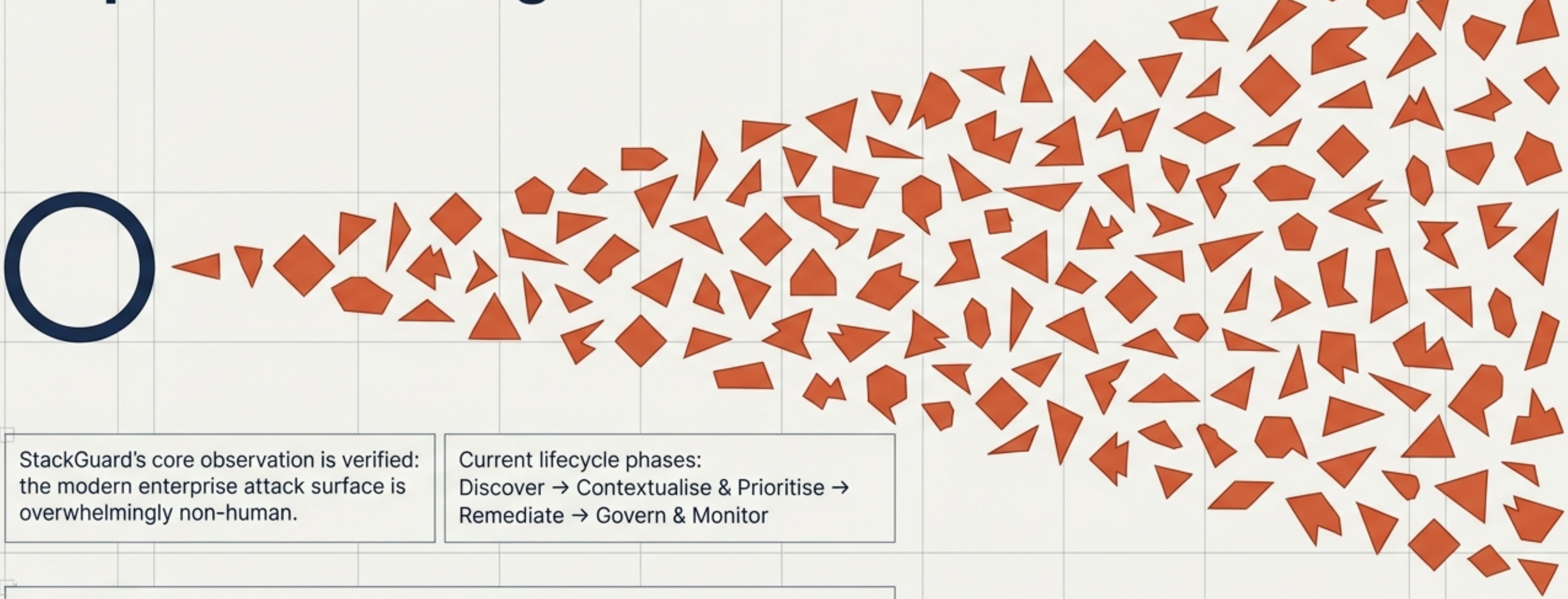
The Cryptographic Substrate for Machine Identity

Mapping the integration of SG/Send PKI architecture with StackGuard NHI Governance (v0.13.12).



The proliferation of non-human identities outpaces human governance 80 to 1



StackGuard's core observation is verified: the modern enterprise attack surface is overwhelmingly non-human.

Current lifecycle phases:
Discover → Contextualise & Prioritise → Remediate → Govern & Monitor

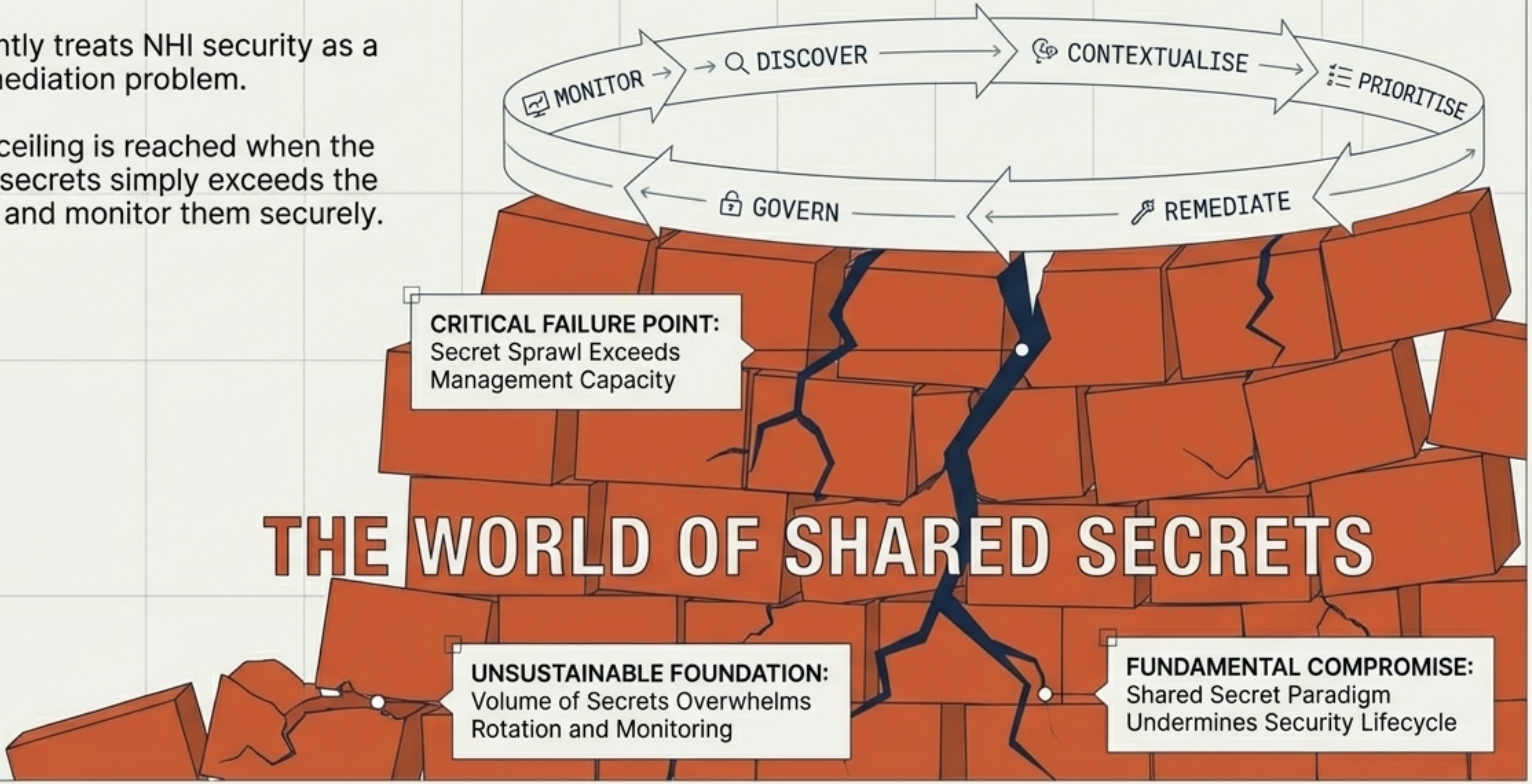
Core Pain Points:

- Lack of NHI inventory
- Zombie identities
- Over-privileged access
- Secret spillage

The architectural ceiling: managing the the fallout of shared secrets

StackGuard currently treats NHI security as a detection and remediation problem.

The architectural ceiling is reached when the volume of shared secrets simply exceeds the capacity to rotate and monitor them securely.



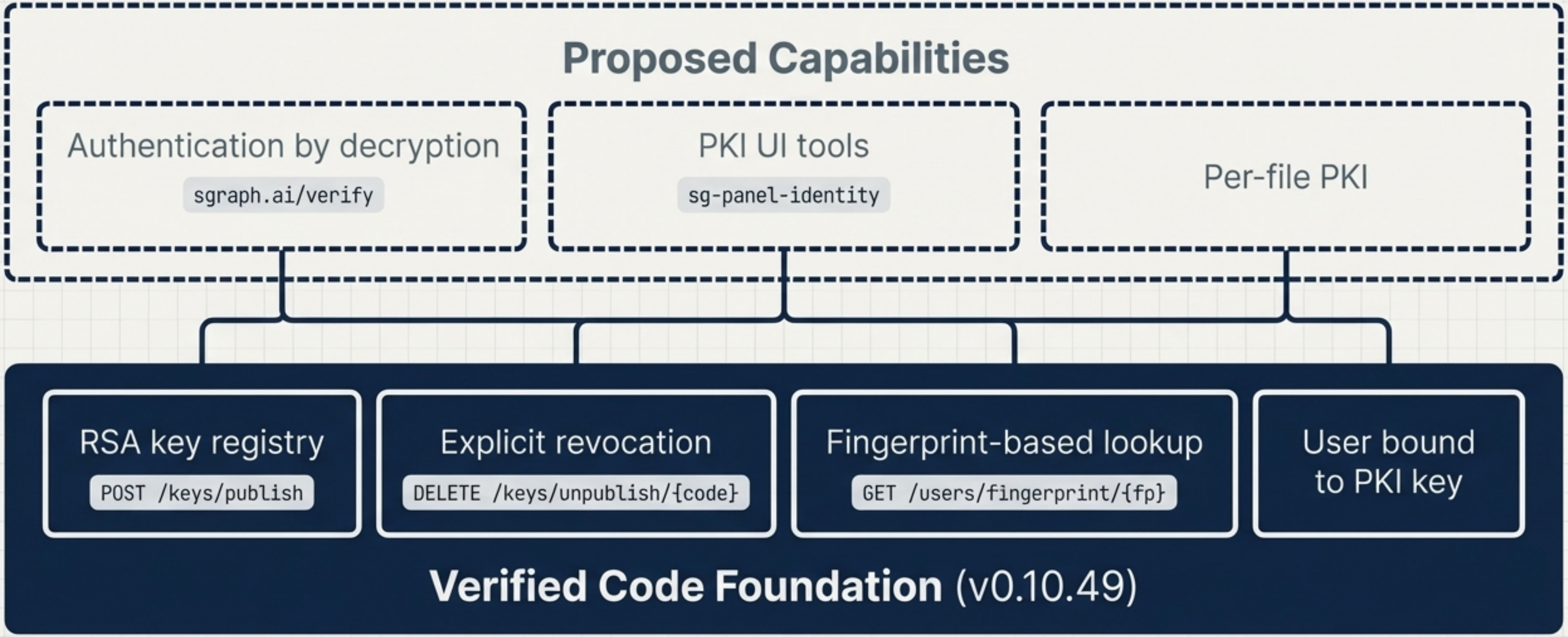
Transitioning from secret remediation to cryptographic identity

	The World of Shared Secrets (Current Status Quo)	The World of Cryptographic Identity (SG/Send PKI)
Identification Inter	Relies on possessing a secret	Identified by a public key fingerprint (**nothing to steal**)
Revocation Inter	Unverifiable, leading to zombie NHIs	Atomic, auditable, and verifiable by any party
Authentication Inter	Requires the transmission of a secret across the wire	Proved via the decryption of a cryptographic challenge

Authentication by decryption eliminates secret spillage entirely

Zero shared secrets on the wire equals zero spillage possible.

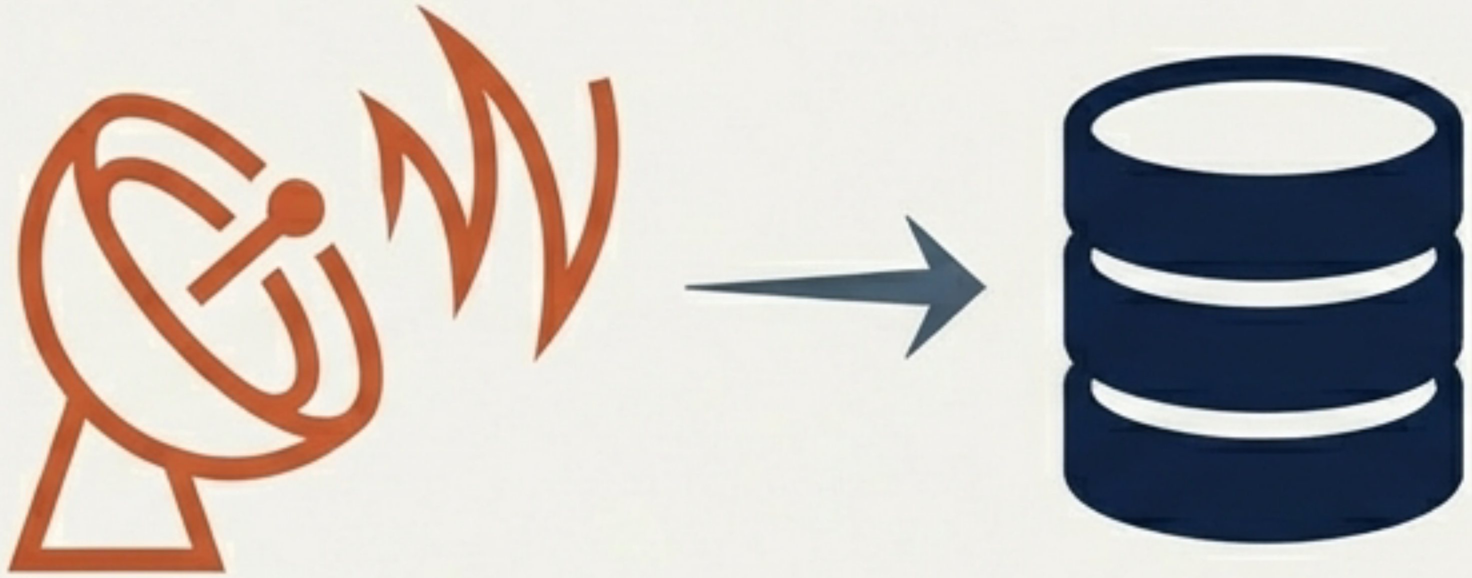
Mapping the SG/Send substrate to the StackGuard pipeline



Angle 1: Cryptographic inventory [redacted] replaces secret scanning

Discovery

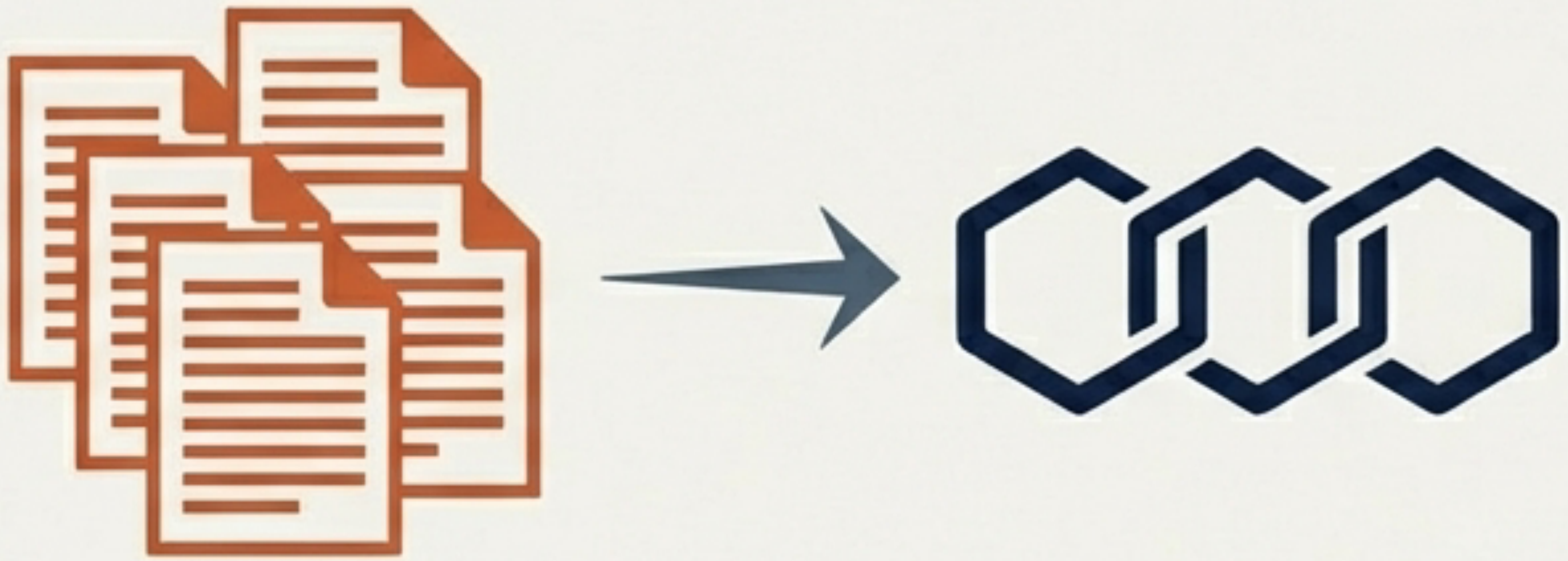
StackGuard's discovery phase upgrades from a reactive secret scan to a definitive key registry query.



`GET /keys/lookup/{code}`

Governance

The SG/Send transparency log provides a tamper-evident audit trail, replacing fragile log scraping with cryptographic proof.

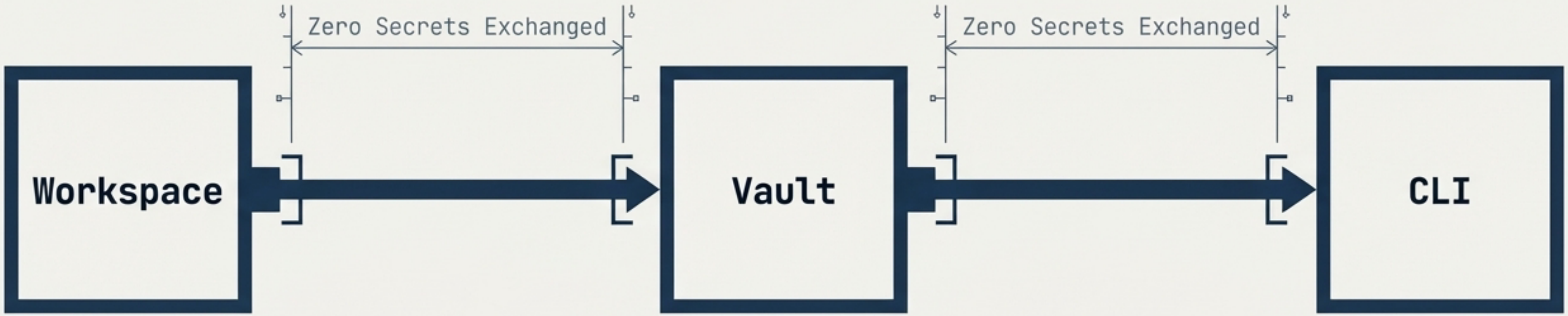


`GET /keys/log`

Angle 2: The zero-shared-secret workflow is already verified

StackGuard's customers want to reduce their NHI attack surface. The current answer is "rotate tokens faster." The ultimate answer is "replace tokens entirely."

SG/Send has already built the working proof-of-concept for cross-system workflows operating with zero shared secrets.



[redacted]

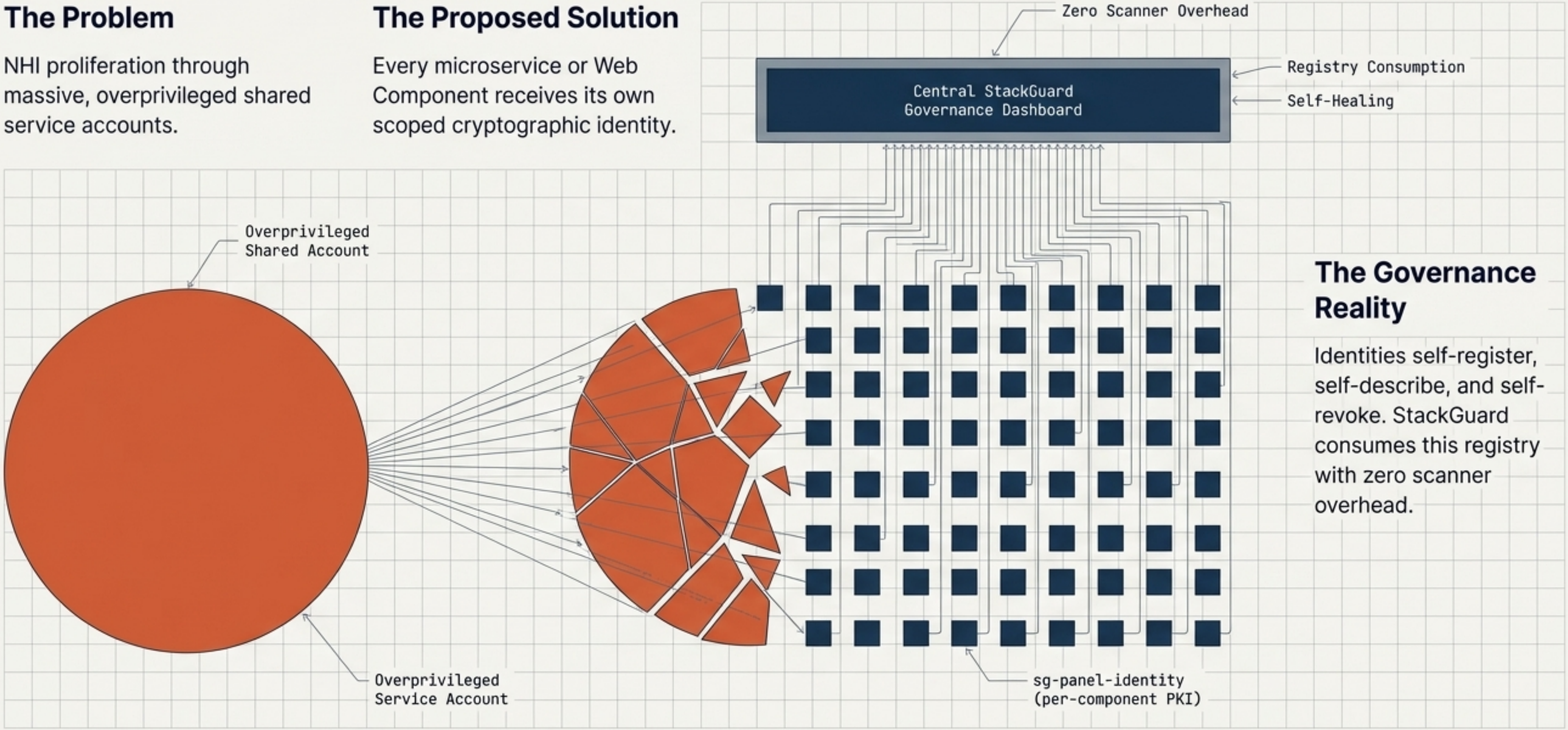
Angle 3: Governing the micro-identity model at enterprise scale

The Problem

NHI proliferation through massive, overprivileged shared service accounts.

The Proposed Solution

Every microservice or Web Component receives its own scoped cryptographic identity.

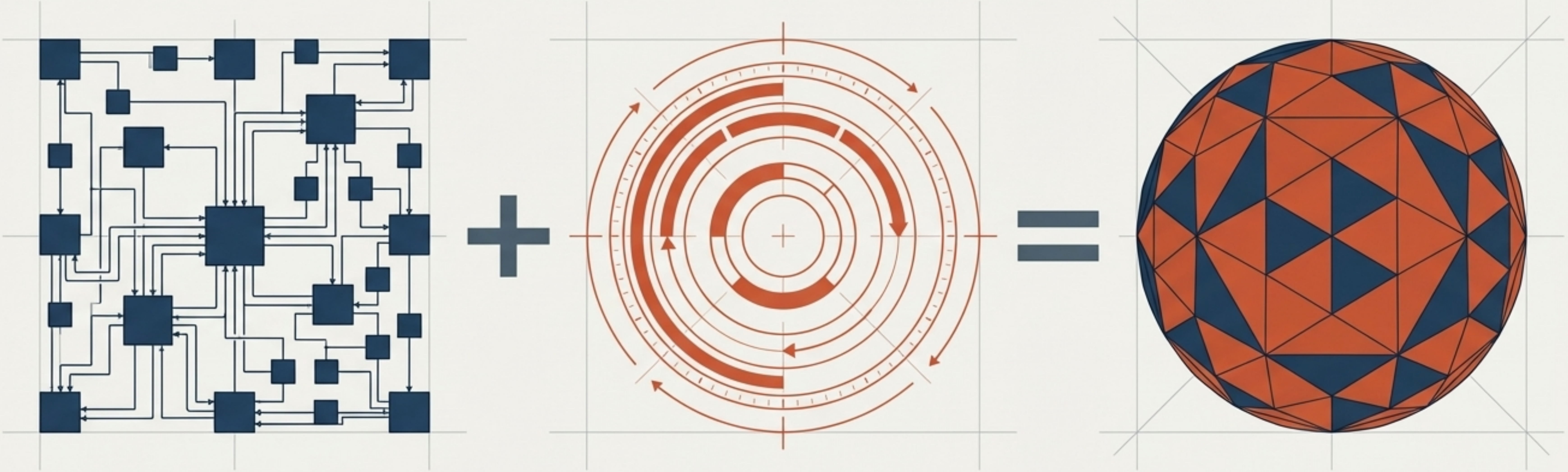


The Governance Reality

Identities self-register, self-describe, and self-revoke. StackGuard consumes this registry with zero scanner overhead.

The Joint Value Equation

SG/Send is not a competitor. It is the architectural substrate that makes StackGuard's ultimate vision possible.



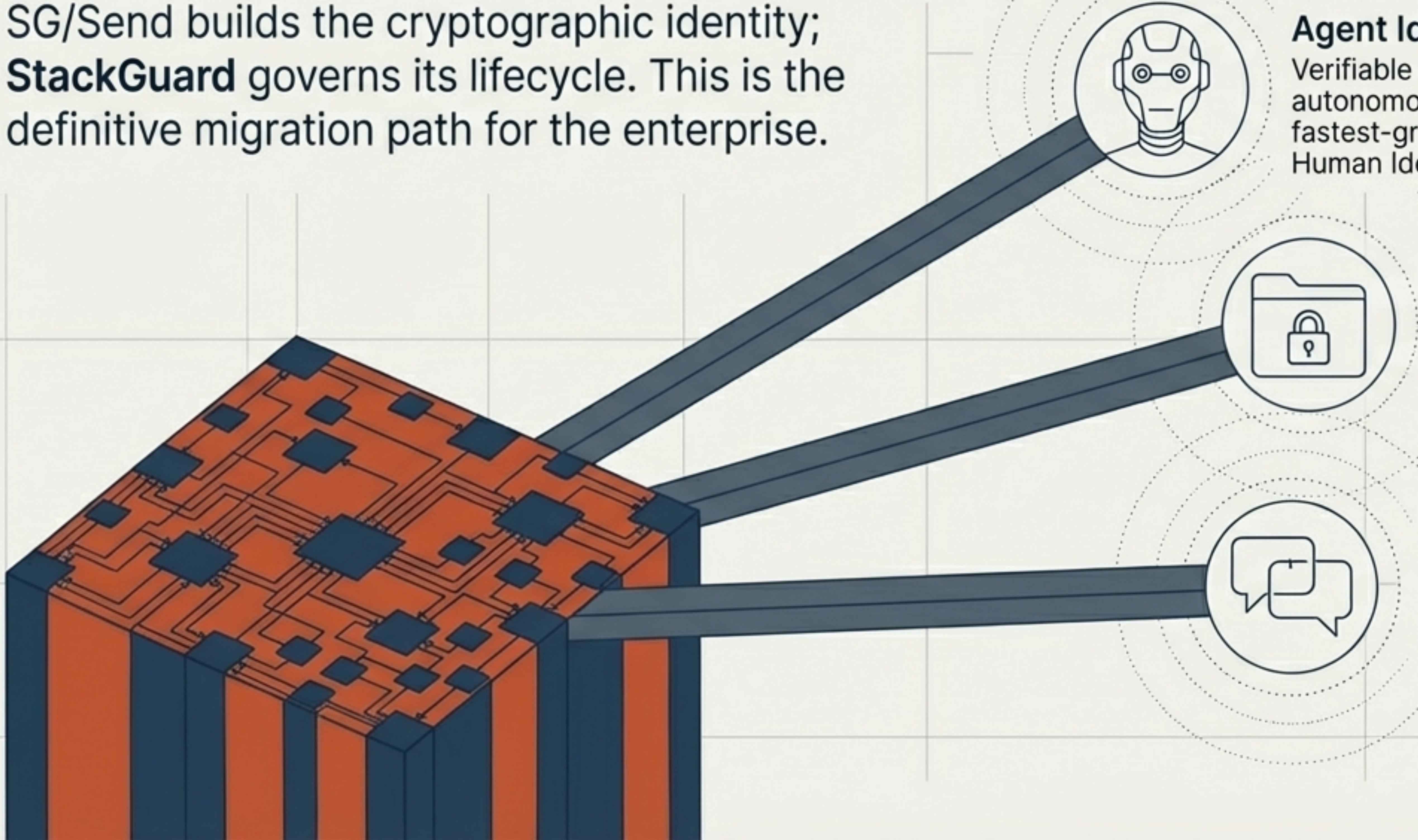
SG/Send: Zero-Knowledge PKI Substrate

StackGuard: NHI Governance Engine

The Future of Proactive Machine Identity

Extending the substrate to Agents, Files, and beyond

SG/Send builds the cryptographic identity; **StackGuard** governs its lifecycle. This is the definitive migration path for the enterprise.



Agent Identity Framework

Verifiable cryptographic identities for autonomous AI. Agents become the fastest-growing category of Non-Human Identity.

Per-File PKI

Verifiable custody chains for compliance artifacts.

Signed Communications

Architecturally impossible service impersonation.